

(b) Using convolution theorem, evaluate $L^{-1} \left\{ \frac{1}{(s^2 + a^2)^2} \right\}$.

(06)

IET, DAVV, Indore
B.E. II (Civil)
Class Test III (Nov-2023)
3AVRC1: Applied Mathematics-III

Time: 70 min

Max Marks: 20

Note: Attempt any FOUR questions. All questions carry equal marks.

Q.1. Evaluate $\int_0^6 \frac{dx}{1+x^2}$ by using (i) Trapezoidal rule (ii) Simpson's 1/3 rule (iii) Weddle's Rules. (05)

Q.2. Calculate the first & second derivatives of the function tabulated below, at the point $x = 1.1$ (05)

x	1.0	1.2	1.4	1.6	1.8	2.0
y	0	0.128	0.544	1.296	2.432	4.000

Q.3. Find the Fourier series of the function $F(x) = x^2$, $-\pi$ to π . (05)

Q.4. Find the Fourier Transform of (05)

$$F(x) = 1 - x^2, |x| \leq 1$$
$$0, |x| \geq 1$$

Hence Evaluate $\int_0^\infty \frac{x \cos x - \sin x}{x^3} \cos \frac{x}{2} dx$.

Q.5. Define Sampling and its types. (05)

BE-II (Electronics & Instrumentation Engg.)
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Q.1 Using Cauchy's Integral formula, evaluate $\int_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz$, where C is the circle $|z| = 3$.

Q.2 Using residue theorem evaluate $\oint_C \frac{12z-7}{(z-1)^2(2z+3)} dz$ where C is the circle (i) $|z| = 2$, (ii) $|z+i| = \sqrt{3}$

Q.3 Obtain Fourier series of the function $f(x) = \begin{cases} -\pi & -\pi \leq x < 0 \\ x & 0 < x \leq \pi \end{cases}$

And hence prove that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$.

Q.4 Find the Fourier half-range cosine series of the function $f(x) = \begin{cases} \pi x & 0 \leq x \leq 1 \\ \pi(2-x) & 1 \leq x \leq 2 \end{cases}$

Q.5 A random variable X has the density function

$$f(x) = \begin{cases} ax, & 0 \leq x < 1 \\ a, & 1 \leq x \leq 2 \\ -ax + 3a, & 2 < x \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

Determine 'a' and compute $P(X \leq 1.5)$.
